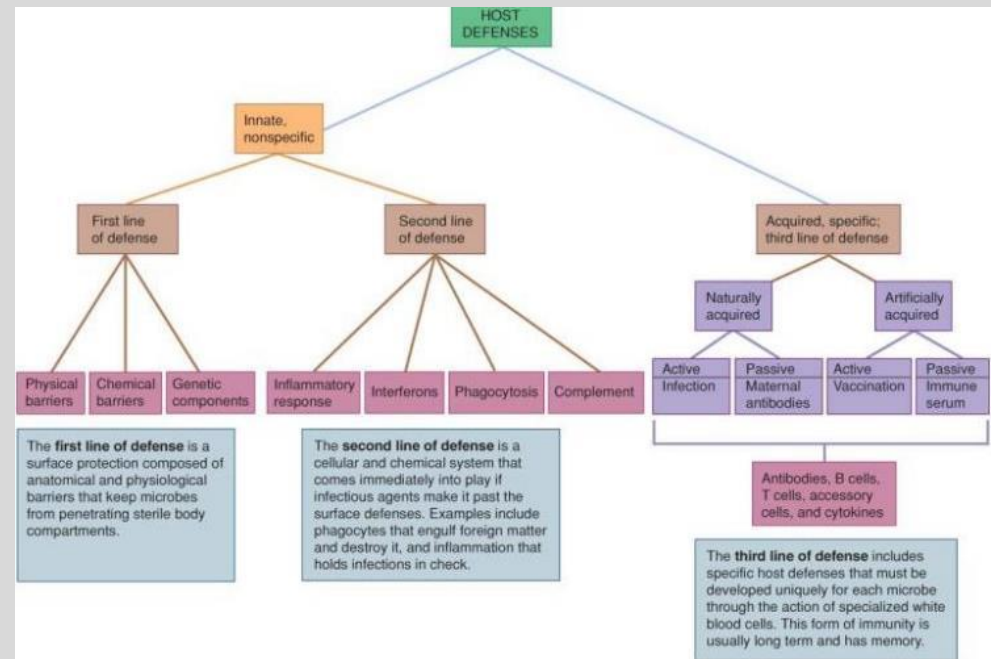


Session Objectives

1. Compare and contrast innate and adaptive IR
2. Classify the innate IR into 1st and 2nd line
3. Distinguish the different components of the 1st and 2nd line of defense
4. Explain key innate processes of inflammation, phagocytosis, fever, interferon & complement action
 - a) Identify the role of cytokines in these processes
 - b) Identify the role of PRRs (pattern recognition receptors) in the inflammatory response
5. Identify deficiencies in innate immunity and the diseases it causes

Innate IR

- First line and second line
- 1st line:
 1. Physical, chemical & biological
 2. skin, ear wax, defensins, tears, mucus, stomach acid, normal flora, bile, etc.
- 2nd line: inflammation, fever, complement, interferon (IFN), phagocytosis, & NK cells



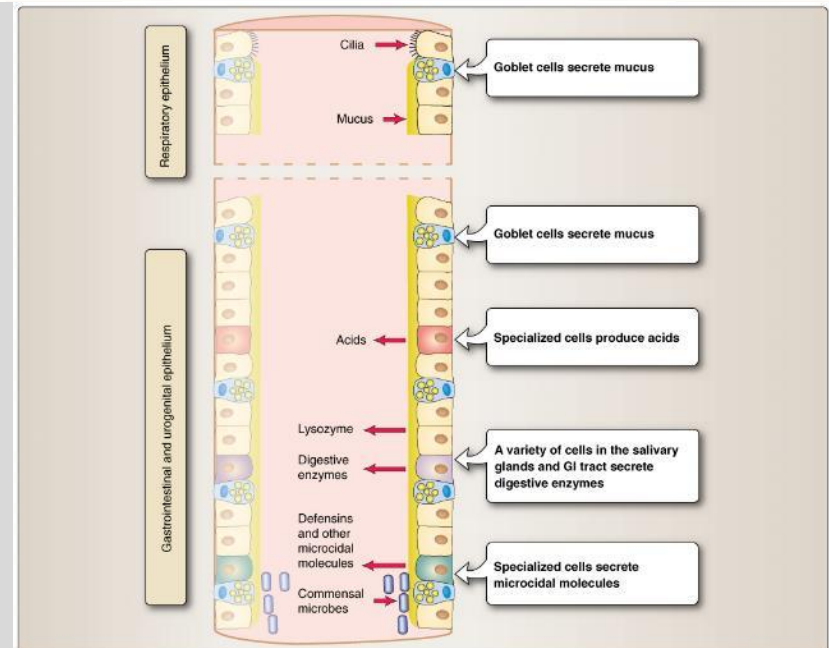
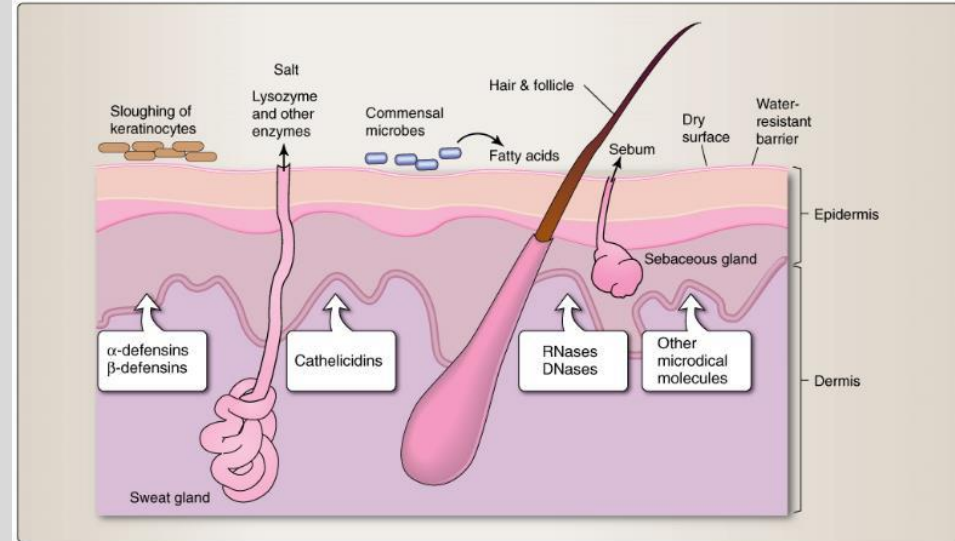
First line: Chemical

- Chemical barriers:

1. pH (skin = 5.5; stomach = 1-3; vagina = 4.5)

2. Microcidal molecules:

- a) Lysozyme
- b) Defensins
- c) Cathelicidins
- d) RNases/DNases
- e) Salt



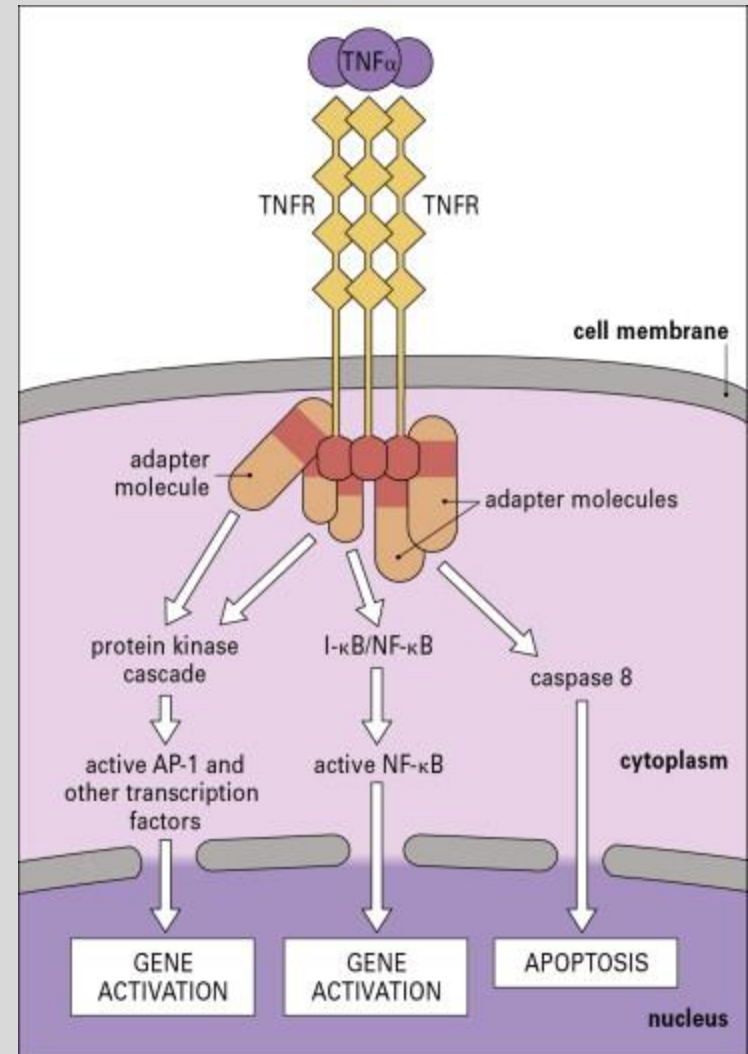
Second line: Inflammation

- A response to tissue damage (physical, infectious, etc.)
- Inflammatory mediators i.e., histamine, C5a, TNF- α
- Cardinal signs: Heat, Pain, Redness, Swelling, (From Latin: *Calor*, *dolor*, *rubor*, and *tumor*), & Loss of Function
- Coordinated by Macrophages



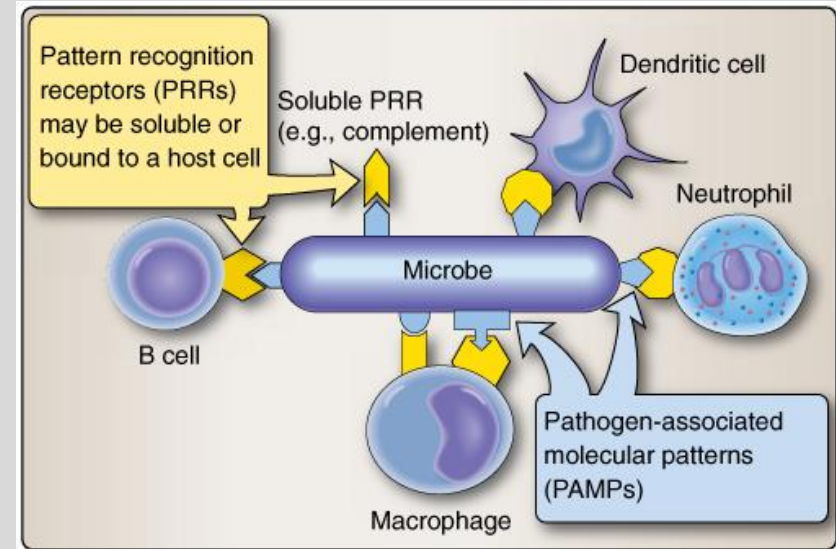
Second line: Inflammation

- Cytokines: $\text{TNF-}\alpha$, IL-1 & $\text{IFN-}\gamma$
- Activate **NF- κ B** (TF)
- NF- κ B master-switch of the immune system
- $\text{TNF-}\alpha$
 - Prothrombotic
 - Leukocyte adhesion & migration
 - Macrophage activation
 - Hematopoiesis and lymphocyte development



Recognition

- Pattern recognition receptors (PRR)
 1. **Secreted** molecules (serum and bodily fluids; e.g., CRP, MBL)
 2. **Receptors** present on the cell surface & on endocytic vesicles (TLR)
 3. **Intra-cytoplasmic** recognition molecules (NOD-like receptors)
- Bind to PAMPs (LPS, PG)
- Coded by genome/no rearrangement/identical in all individuals



Source: Thao Doan, Roger Melvold, Susan Viselli and Carl Waltenbaugh. 2013. Lippincott's Illustrated Reviews: Immunology, 2ed. Lippincott Williams & Wilkins publisher

TLRs

- Found in most tissues (10 TLR)
- Activate phagocytes → APCs
- Activate NF-κB
- TLR3 deficiency susceptibility to HSV
- TLR7 higher viral load & disease progression in HIV

Receptor	Location	Ligand	Pathogen
TLR1	Cell surface	Lipopeptides	Gram -negative bacteria mycobacteria
TLR1/TLR2	Cell surface	Tri-acyl lipoprotein	Bacteria
TLR2	Cell surface	Lipoteichoic acid Lipoarabinomannan Zymosan Glycoinositol-phospholipids	Gram-positive bacteria Mycobacteria Fungi T. Cruzi
TLR2/TLR6	Cell surface	Di-acyl lipoprotein	Bacteria
TLR3	Endosome or Cell surface	dsRNA	Viruses
TLR4	Cell surface	LPS	Gram -negative bacteria
TLR5	Cell surface	flagellin	Bacteria
TLR6	Cell surface	Di-acyl lipopeptides	Mycobacteria
TLR7	Endosome	ssRNA	Viruses
TLR8	Endosome	ssRNA	Viruses
TLR9	Endosome	Unmethylated CpG DNA	Bacteria

David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher

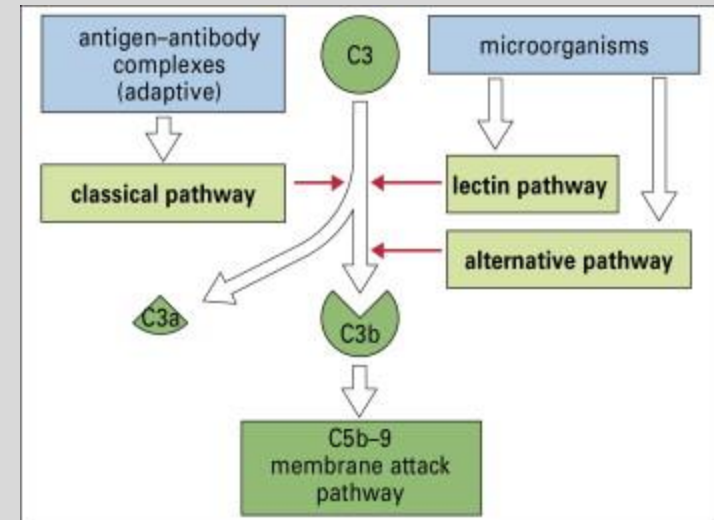
Second line: Complement

~16 soluble serum proteins
(10%)

Inactive; ancient

Activated

- Classical pathway (adaptive IR)
- Alternative pathway (innate IR)
- Mannan-binding lectin (MBL) pathway (innate IR)

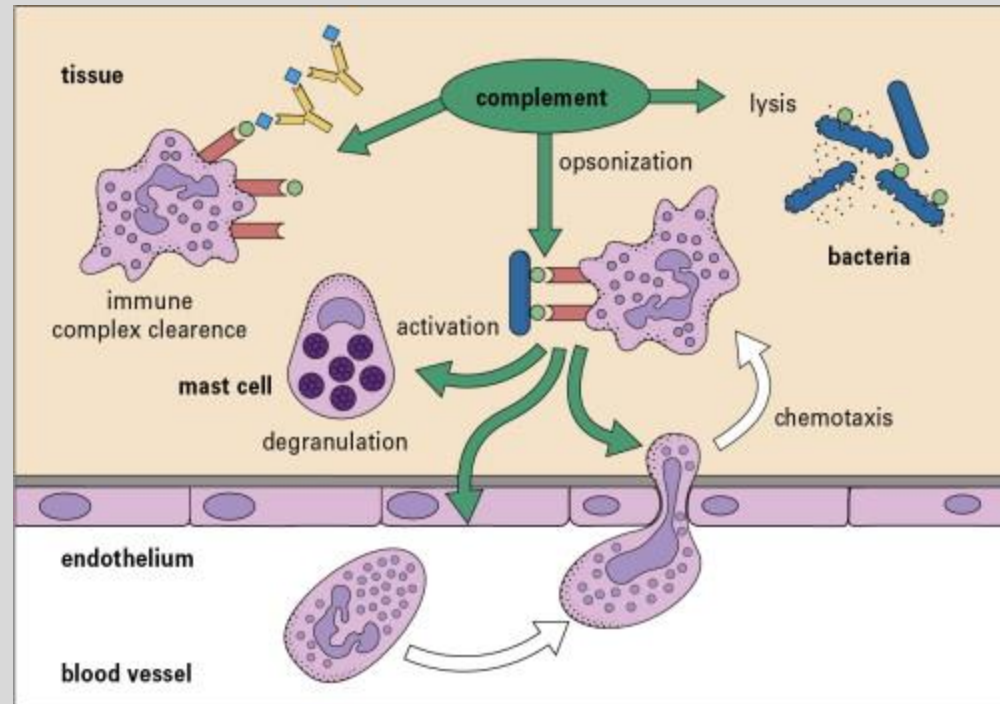


David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher.

Second line: Complement function

- Function:

1. **Opsonization** (*C3b* & *C4b*)
2. Lysis of bacteria and enveloped viruses (*MAC*)
3. Inflammation & chemotaxis (*C3a* & *C5a*)
4. Efficient development of Ab response (*priming of B-cells*)
5. Clearance of immune complexes and apoptotic cells



David Male, Jonathan Brostoff, David Roth, and Ivan Roitt. 2013. *Immunology* 8th edition. Elsevier publisher

Second line: Complement (activation)

- Classical: Ab (IgM/IgG) bound to antigen and requires Ca^{++}
- Lectin: microbial carbohydrates (mannose & NAG)
- Alternative:
 - Constant state of low-level activation
 - Requires Mg^{++} & factor B
 - Activated by surface molecules (i.e., LPS)

Chemokines & their receptors

- Both ligand/receptors vary resulting in selection of leukocytes entering tissues
- Defect: leukocyte adhesion deficiency (LAD) syndrome (β -integrins)
- Therapeutic application: Abs against VLA-4 (the ligand for VCAM-1) MS patients (limit migration of T-cells into the CNS)

	Th1		MØ		Th2		eosinophil	basophil	neutrophil
	(CCR1)	CCR1	CCR2	CCR2	CCR5 → CCR5	CCR3 → CCR3	CCR3 → CCR3	CCR3 → CCR3	(CCR1)
	CCR2	CCR2	CCR5	CCR5	CCR3	CCR3	CCR3	CCR3	CXCR1
	CXCR3								CXCR2
CCL3	+	+				+	+		+
CCL4	+	+							
CCL5	+	+			+	+	+		+
CCL2	+	+			+	+	+		
CCL7,8	+	+			+	+	+		+
CCL13	+	+			+	+	+		
CCL11					+	+	+		
CXCL8									+
CXCL1,2,3									+
CXCL9,10,11	+								

Second Line: Phagocytes (intracellular receptors)

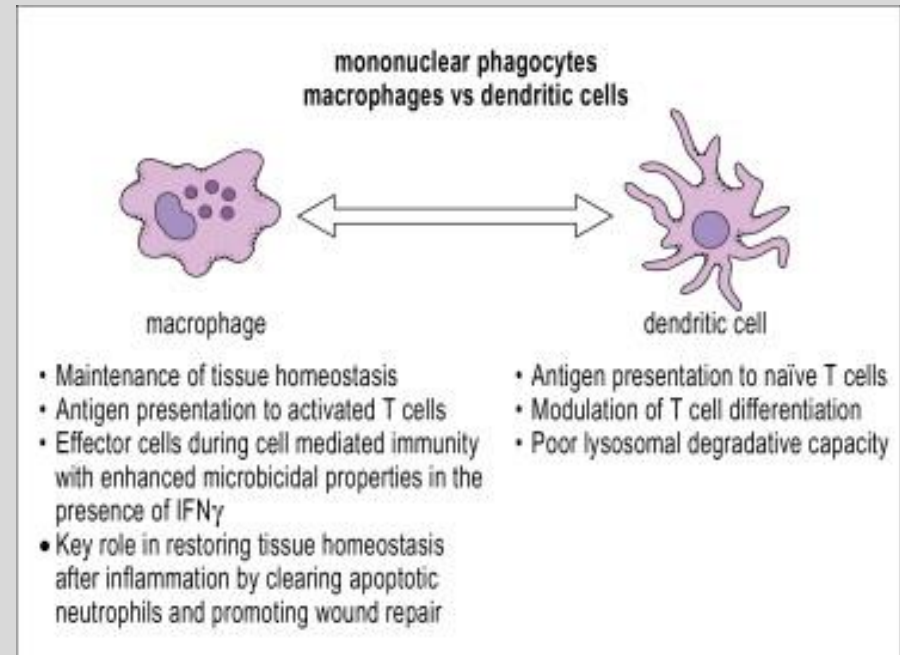
- Cytosolic receptors recognize intracellular pathogens (part of PRRs)
 - a) NLR (*NOD-like receptors*) - PG
 - b) RLHs (*Retinoic acid-inducible gene I-like helicases*) - dsRNA
- NLRs form the "**inflammasome**" triggers inflammatory cell death (pyroptosis) of infected cells
- NOD polymorphisms:
 1. Atopic eczema
 2. Asthma
 3. Increase IgE
 4. Increase risk of Crohn's disease
 5. Increase in bacterial infections

Second Line: Phagocytes (Neutrophils vs Macrophages)

- Key differences between Neutrophils & Macrophages
 - a) Neutrophils:
 - i. Short lived
 - ii. Acute response
 - iii. Fast clearance
 - iv. Store a wide range of antimicrobial polypeptides in intracellular granules (~50% defensins)
 - v. Readily produce oxygen radicals
 - b) MΦ:
 - i. Long lived
 - ii. Chronic inflammation/infection
 - iii. Less microbicidal and cytotoxic
 - iv. Recruit neutrophils
 - v. Requires activation by PRRs
 - vi. Display a higher degree of specificity which makes them better suited as sentinels of the immune system

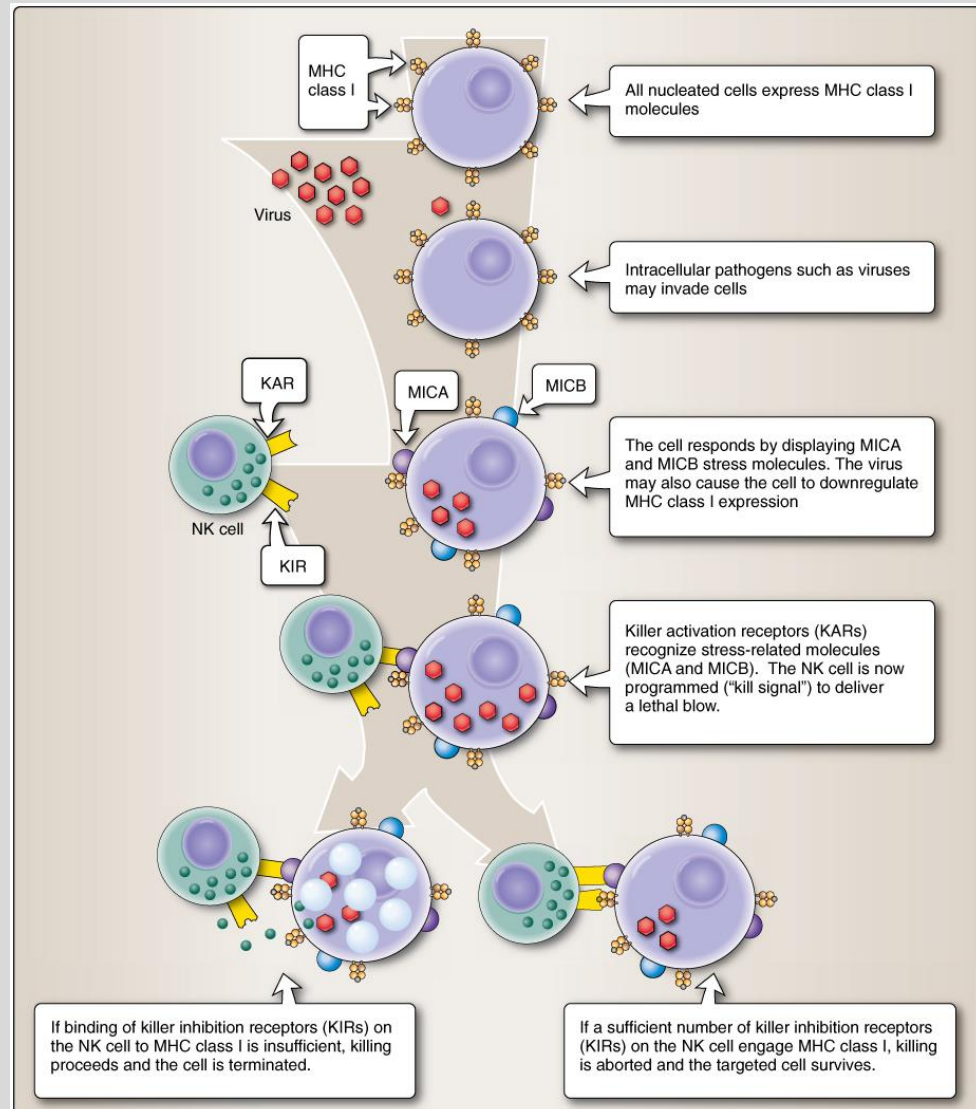
Second line: Phagocytes' cytokines

- Secrete:
 - a) IL-1 (pyrogen)
 - b) TNF- α (inflammation)
 - c) IL-8 & IL-12 (attract neutrophils & NK cells)



Second line: NK cells

- NK cells have killer activation (KAR) & killer inhibition receptors (KIR)
- KAR recognize stress-associated molecules (MICA, MICB)
- NK cells use killer inhibition receptors (KIR) to assess MHC I (insufficient levels of KIR-MHC I → killing target host cell)
- Viruses & malignant events depress MHC I



Primary Immunodeficiencies in innate IR

Phagocyte disorders:

1. *Chediak-Higashi syndrome*: defect in phagosome-lysosome fusion
2. *Chronic Granulomatous disease*: defect in microbicidal enzymes/O₂ radicals
3. *IFN- γ receptor* deficiency
4. *Leukocyte adhesion deficiency* (LAD-1/LAD-2)

Primary Immunodeficiencies in innate IR

Complement disorders:

1. Defects in **alternative** pathway & **MBL** lead to increase susceptibility to infections
2. Defects in **classical** pathway not associated with significant increase in susceptibility to infections (ex. C3 deficiency & *encapsulated*)
3. Deficiencies in C1, C2 & C4 → inefficient clearance of immune complexes (increase in type III hypersensitivity disease)
4. Deficiency in C regulation (C1 inhibitor = hereditary angioedema)

Summary Slide

- Compared innate IR to adaptive IR
- Classified innate IR into 1st and 2nd line of defense
- Discussed role of the 1st line of defense (physical and chemical)
- Discussed role of the 2nd line of defense (inflammation, complement, phagocytosis, etc.)
- Pointed out deficiencies in these systems and how that leads to diseases or syndromes